

Reviewer Pack — Minimal Rank of Twisted Derivatives in Noncommutative Geomet...

Entry 7f79adc81bff · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Twisted Derivatives in Noncommutative Geometry vs. Communication Complexity for Disjointness

Entry ID: 7f79adc81bff **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 20:37:18 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry **Field B** (complexity object): Communication Complexity

Statement:

[‘The minimal rank of the twisted derivatives of a matrix algebra over a skew field, when applied to the disjointness function, provides a lower bound on the randomized communication complexity for disjointness.’, ‘Specifically, for all $n \leq 40$ instances, if the minimal rank of the twisted derivative is at least $\Omega(n)$, then the randomized communication complexity for disjointness is also $\Omega(n)$.’, ‘Conversely, if there exists an instance with minimal rank less than $\Omega(n)$ and communication complexity greater than or equal to $\Omega(n)$, the conjecture is falsified.’]

Rationale (proposer’s reasoning):

[‘Noncommutative geometry has been successfully applied to quantum information theory and may provide new insights into classical communication complexity problems.’, ‘The twisted derivatives in noncommutative geometry capture the essence of non-locality, which is a central feature of communication complexity for disjointness.’, ‘This conjecture aims to bridge the gap between noncommutative geometry and communication complexity, potentially leading to new algorithms or lower bounds.’]

Taxonomy category: COMM_DISJ (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b186f4e029ec05fd

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if for all $n \leq 40$ instances, the mean minimal rank of the twisted derivatives is at least $\Omega(n)$ and no instance has a minimal rank less than $\Omega(n)$ with communication complexity greater than or equal to $\Omega(n)$. The conjecture is falsified if any instance violates this criterion.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal rank twisted derivatives noncommutative geometry AND communication complexity disjointness - Disjointness communication complexity lower bound minimal rank twisted derivative skew field - Twisted derivative matrix algebra skew field disjointness function communication complexity randomized complexity

Top relevant hits considered: - [<http://arxiv.org/abs/1209.3595v2>] Noncommutative complex differential geometry - [<http://arxiv.org/abs/1612.06688v2>] The Ricci Curvature in Noncommutative Geometry - [<http://arxiv.org/abs/1007.0885v1>] Noncommutative spaces with twisted symmetries and second quantization - [<http://arxiv.org/abs/1806.02734v3>] Spectral lower bounds for the orthogonal and projective ranks of a graph - [<http://arxiv.org/abs/1403.8106v1>] Recent advances on the log-rank conjecture in communication complexity - [<http://arxiv.org/abs/cs/0111062v2>] One-way communication complexity and the Neciporuk lower bound on formula size - [<http://arxiv.org/abs/0911.3482v5>] Complexity of Networks (reprise) - [<http://arxiv.org/abs/quant-ph/0405018v2>] Improved Bounds on the Randomized and Quantum Complexity of Initial-Value Problems

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_disjointness_instance(n):
        A = [random.randint(0, 1) for _ in range(n)]
        B = [random.randint(0, 1) for _ in range(n)]
        return A, B

    def matrix_multiplication(A, B):
        n = len(A)
        C = [[0] * n for _ in range(n)]
        for i in range(n):
            for j in range(n):
                for k in range(n):
                    C[i][j] += A[i][k] * B[k][j]
```

```

    return C

def gaussian_elimination(A, b):
    n = len(A)
    for i in range(n):
        max_row = i
        for j in range(i + 1, n):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]
        b[i], b[max_row] = b[max_row], b[i]
        for j in range(i + 1, n):
            factor = A[j][i] / A[i][i]
            for k in range(n):
                A[j][k] -= factor * A[i][k]
            b[j] -= factor * b[i]
    x = [0] * n
    for i in range(n - 1, -1, -1):
        x[i] = (b[i] - sum(A[i][j] * x[j] for j in range(i + 1, n))) / A[i][i]
    return x

def calculate_minimal_rank(A):
    n = len(A)
    identity = [[1 if i == j else 0 for j in range(n)] for i in range(n)]
    augmented_matrix = [A[i] + identity[i] for i in range(n)]
    rank = 0
    for row in gaussian_elimination(augmented_matrix, [0] * n):
        if any(row[j] != 0 for j in range(n)):
            rank += 1
    return rank

def communication_complexity(A, B):
    n = len(A)
    count = 0
    for i in range(n):
        if A[i] != B[i]:
            count += 1
    return count

results = []
for _ in range(30): # Ensure at least 30 instances per seed
    n = random.choice([5, 10, 15, 20, 30, 40])
    A, B = generate_disjointness_instance(n)
    rank_A = calculate_minimal_rank(A)
    comm_complexity = communication_complexity(A, B)

    results.append({
        "metric_name": "Minimal Rank of Twisted Derivatives",
        "metric_value": rank_A,
        "instances_tested": 1,
        "conjecture_holds": rank_A >= n,
        "counterexample": f"n={n}, rank_A={rank_A}" if not rank_A >= n else ""
    })

return {
    "seed": seed,

```

```

    "metric_name": "Minimal Rank of Twisted Derivatives",
    "metric_value": sum(r["metric_value"] for r in results) / len(results),
    "instances_tested": sum(r["instances_tested"] for r in results),
    "conjecture_holds": all(r["conjecture_holds"] for r in results),
    "counterexample": next((r["counterexample"] for r in results if not r["conjecture_holds"]))
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3f774267.py", line 102, in <module>
    result = run_trial(seed)
              ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3f774267.py", line 76, in run_trial
    rank_A = calculate_minimal_rank(A)
              ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3f774267.py", line 57, in calculate_minimal_rank
    augmented_matrix = [A[i] + identity[i] for i in range(n)]
                        ~~~~~^~~~~~
TypeError: unsupported operand type(s) for +: 'int' and 'list'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying the conjecture’s support or falsification. | next: Investigate and fix the error in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14767
2	preregistration	ollama_remote	glm4:latest	0	0	9516
3	novelty	ollama_remote	glm4:latest	0	0	8754
4	novelty	ollama_remote	glm4:latest	0	0	9761
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16842
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11053
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13995
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12926
9	judge	ollama_remote	glm4:latest	0	0	11816

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 109431 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/7f79adc81bff.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/7f79adc81bff.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/7f79adc81bff.tar.gz (if generated)